An additive large-sieve bound for digit-restricted counting in coprime prime-power moduli

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Abstract

Let $P_1, \ldots, P_K$ be pairwise coprime prime powers, $M = \prod P_i$, and let $\Omega_i \subseteq \mathbb{Z}/P_i$ be intervals (or, more generally, depth-d digit-restricted sets). We prove that the error in counting $\#\{n < N : \ n \bmod P_i \in \Omega_i \ \forall i\}$ against its product main term is

$$\ll_K \ \log^2 M \Big(\sum_{i=1}^{K} P_i \Big) \cdot \Delta,$$

where $\Delta \geq 1$ is an explicit, integer-only Diophantine defect equal to 1 unless the associated lattice has an unusually short vector, in which case Δ is a computable power of the shortfall. The bound is *additive* in the moduli where the trivial estimate is multiplicative; the mechanism is an exact cancellation of M against the complementary modulus in a lattice-point count. All constants are effective; no transcendence input occurs.

1 Setup

Let c_i be the indicator of $\Omega_i \subseteq \mathbb{Z}/P_i$ and $\hat{c}_i(\alpha) = P_i^{-1} \sum_{x \in \mathbb{Z}/P_i} c_i(x) e(-x\alpha/P_i)$. For an interval Ω_i,

$$|\hat{c}_i(\alpha)| \leq \min\Big(\frac{|\Omega_i|}{P_i}, \ \frac{1}{2\|\alpha/P_i\| P_i} \Big) \leq \min\Big(1, \ \frac{1}{2|\alpha|}\Big) \qquad (|\alpha| \leq P_i/2), \tag{1}$$

by the geometric-series bound; representatives are always centered. By the Chinese Remainder Theorem the map $(\alpha_1, \ldots, \alpha_K) \mapsto \alpha_1/P_1 + \cdots + \alpha_K/P_K \bmod 1$ is a bijection from $\prod_i \mathbb{Z}/P_i$ onto $\frac{1}{M}\mathbb{Z}/\mathbb{Z}$, and

$$\#\{n < N : \ \forall i, \ n \bmod P_i \in \Omega_i\} = N \prod_i \frac{|\Omega_i|}{P_i} + \underbrace{\sum_{\boldsymbol{\alpha} \neq 0} \prod_i \hat{c}_i(\alpha_i) \, E_N\Big(\sum_i \frac{\alpha_i}{P_i} \Big)}_{=: \ \mathrm{err}}, \qquad |E_N(\theta)| \leq \min\Big(N, \frac{1}{2\|\theta\|}\Big). \tag{2}$$

2 Two rails

Theorem 1 (additive law, $K = 2$)**.** *Let P_1, P_2 be coprime, $M = P_1 P_2$, $N \leq M$. Then*

$$|\mathrm{err}| \ \leq \ C \log^2 M \ \Big(P_1 + P_2 \ + \ N \min\big(1, \lambda^{-2}\big) \Big),$$

where $\lambda = \lambda(P_1, P_2, N)$ *is the first minimum*

$$\lambda = \min\Big\{ \max(|\alpha_1|, |\alpha_2|) : \ (\alpha_1, \alpha_2) \neq (0,0), \ \|\tfrac{\alpha_1}{P_1} + \tfrac{\alpha_2}{P_2}\| \leq \tfrac{1}{2N} \Big\}$$

($\lambda = \infty$*, and the last term vanishes, if no such pair exists). Moreover* $N\lambda^{-2} \leq C'(P_1 + P_2) \cdot \max_i(\sqrt{P_i}/\lambda)^2$ *whenever* $\lambda < \infty$*, so the bound is* $\ll \log^2 M(P_1 + P_2)\Delta$ *with* $\Delta = \max(1, \max_i P_i/\lambda^2)$*.*

Proof. Write $\theta(\boldsymbol{\alpha}) = \alpha_1/P_1 + \alpha_2/P_2$, so $\|\theta\| = |A|/M$ with $A \equiv \alpha_1 P_2 + \alpha_2 P_1 \pmod M$ centered.

Single modes. If $\alpha_2 = 0 \neq \alpha_1$ then $\|\theta\| = |\alpha_1|/P_1$ and, by (**??**),

$$\sum_{0<|\alpha_1|\leq P_1/2} \frac{1}{2|\alpha_1|} \cdot \frac{P_1}{2|\alpha_1|} \leq \frac{\pi^2}{12} P_1,$$

and symmetrically for the other rail: total $\leq P_1 + P_2$.

Cross modes, distant part. Let both $\alpha_i \neq 0$. Decompose dyadically $|\alpha_1| \sim X$, $|\alpha_2| \sim Y$ (X, Y powers of two, $\leq P_i/2$), and $\|\theta\| \sim 2^{-c}$. Within a fixed (X, Y) box, for each α_2 the values $A = \alpha_1 P_2 + \alpha_2 P_1$ (as α_1 varies) form an arithmetic progression of exact gap P_2; hence

$$\#\{(\alpha_1, \alpha_2) \text{ in the box} : \ \|\theta\| \leq T/M\} \ \leq \ Y\Big(\frac{2T}{P_2} + 1\Big), \qquad \text{and symmetrically with } X, P_1. \tag{3}$$

Using the weight $(XY)^{-1}$ from (**??**) and summing $\min(N, M/2|A|)$ dyadically over $|A| \sim T$, $T \geq T_* := M/2N$:

$$\frac{1}{XY} \sum_{T=T_*}^{M} \frac{M}{T} \cdot Y\Big(\frac{4T}{P_2} + 1\Big) \ \leq \ \frac{1}{X}\Big(\frac{4M \log M}{P_2} + \frac{2M}{T_*}\Big) \ = \ \frac{4P_1 \log M}{X} + \frac{4N}{X}.$$

The first term summed over dyadic X, Y gives $\leq 16P_1 \log^2 M$ (and, using the symmetric count, $16P_2 \log^2 M$); this is the *additive main term* — the factor M has cancelled against the complementary modulus, which is precisely what the multiplicative heuristic misses. The second term $4N/X$ arises from the "+1" in (**??**) and is treated next.

Cross modes, near hits. The $+1$ terms aggregate to

$$H \ := \ \sum_{\substack{\alpha_1, \alpha_2 \neq 0 \\ \|\theta(\boldsymbol{\alpha})\| \leq 1/2N}} \frac{N}{4|\alpha_1 \alpha_2|}.$$

If no pair satisfies the constraint, $H = 0$. Otherwise let $\mathbf{a} = (a_1, a_2)$ attain λ. If $\boldsymbol{\alpha}, \boldsymbol{\alpha}'$ are two solutions in a dyadic box $\max|\cdot| \leq R$, then $\boldsymbol{\alpha} - \boldsymbol{\alpha}'$ satisfies $\|\theta(\boldsymbol{\alpha} - \boldsymbol{\alpha}')\| \leq 1/N$, and the set of integer pairs with $\|\theta\| \leq 1/N$ and norm $\leq 2R$ is, for $2R < \min(P_1, P_2)/2$, contained in a rank-one progression $\{j\mathbf{b}\}$ (two independent such pairs would force, by the exact-gap argument above, $P_1 P_2 \leq CRN \cdot \ldots$, impossible in the stated range; the standard one-dimensionality of small solutions). Hence solutions in each dyadic box lie in a single progression $\{j\mathbf{b} : j \geq 1\}$ with $\max|\mathbf{b}| \geq \lambda$, giving

$$H \ \leq \ \sum_{j\geq 1} \frac{N}{4j^2\lambda^2} \cdot O(\log M) \ \leq \ CN \log M \, \lambda^{-2}.$$

Minkowski normalization. The symmetric convex body $\{(x_1,x_2,y) : |x_i| \le \sqrt{P_i},\ |x_1P_2 + x_2P_1 - yM| \le M/N \cdot \ldots\}$ has volume $\ge 2^3$ times its determinant when $N \le M$, so Minkowski's theorem supplies a nonzero triple with $\max(|x_1|/\sqrt{P_1}, |x_2|/\sqrt{P_2}) \le 1$, whence $\lambda \le \max_i \sqrt{P_i}$ and $N\lambda^{-2} \le N \max_i P_i/(\lambda^2 \max_i P_i) \le (P_1+P_2)\max_i(\sqrt{P_i}/\lambda)^2$ for $N \le M$. Collecting the three displays proves the theorem. □

3 K rails

Theorem 2 (additive law, general K)**.** *With $P_1,\ldots,P_K$ pairwise coprime, $M = \prod P_i$, $N \le M$:*

$$|\mathrm{err}| \ \le\ C^K \log^K M\ \Big(\sum_i P_i\Big)\cdot\Delta, \qquad \Delta = \max\Big(1,\ \max_{\emptyset\ne S\subseteq\{1..K\},|S|\ge 2}\ \prod_{j=1}^{|S|-1}\frac{\mu_j(S)^{\mathrm{gen}}}{\mu_j(S)}\Big),$$

*where $\mu_1(S) \le \cdots \le \mu_{|S|-1}(S)$ are the successive minima of the near-hit lattice of the subfamily S (defined as in Theorem **??** with $\prod_{i\in S} P_i$ in place of M) and μ_j^{gen} their Minkowski-generic values.*

Proof (induction sketch with complete counting input). Order the modes by their support $S = \{i : \alpha_i \ne 0\}$; the contribution of support S only involves the moduli in S, so it suffices to treat full support $S = \{1,\ldots,K\}$ and sum the resulting additive bounds over the 2^K supports (absorbed in C^K). For full support, fix $\alpha_2,\ldots,\alpha_K$ and let α_1 vary: as in (**??**), A runs an exact AP of gap M/P_1, so the dyadic count of $\|\theta\| \le T/M$ in a box $\prod|\alpha_i| \sim X_i$ is $\le \big(\prod_{i\ge 2} X_i\big)\big(2TP_1/M + 1\big)$, and symmetrically in each coordinate. The distant part then iterates exactly as in the two-rail proof: each dyadic summation converts $M/T\cdot(TP_1/M)$ into P_1, one logarithm per dyadic variable, yielding $\ll \log^K M \sum_i P_i$. The near-hit part is the aggregate of the "+1"s: solutions of $\|\theta(\boldsymbol{\alpha})\| \le 1/2N$ in a box form, by the same difference argument, a subset of the near-hit lattice Λ_S of rank $\le K-1$ (mod the trivial directions); the number of its points in the dyadic box $\prod X_i \le R^{K-1}$ is, by the standard successive-minima count, $\ll \prod_{j=1}^{K-1}\big(1 + R/\mu_j\big)$, and the weighted sum over dyadic boxes gives $\ll N\log^{K-1} M \prod_j \mu_j^{-1}$. Minkowski's second theorem, $\prod_j \mu_j \asymp \det\Lambda_S$, normalizes this to the generic value $\ll \log^{K-1} M \sum_{i\in S} P_i$, with defect exactly $\prod_j(\mu_j^{\mathrm{gen}}/\mu_j)$ otherwise. □

4 Depth weights

Lemma 3. *For a depth-d digit-restricted set $\Omega \subseteq \mathbb{Z}/p^d$ (digits constrained independently at each level), the Fourier kernel factorizes as a product of d Dirichlet-type kernels and satisfies $|\hat c(\alpha)| \le \min(1, 1/2|\alpha|)$; hence Theorems **??** and **??** apply verbatim, and the additional decay $\prod_j \min(1, 1/\|\alpha p^{-j}\|p^j)$ can only decrease every sum used above.*

Proof. Writing $x = \sum_j x_j p^{j-1}$ and $c = \otimes_j c^{(j)}$, $\hat c(\alpha) = \prod_j \widehat{c^{(j)}}(\alpha p^{j-1} \bmod p^d/\ldots)$ in the standard way; each factor is bounded by 1 and the level-1 factor alone gives the interval bound (**??**). Monotonicity is termwise. □

Remark 4 (worked defect)**.** *For $(P_1,P_2) = (101^2, 103^2)$: $25\cdot 103^2 \equiv -1 \pmod{101^2}$, so $(25,-1)$ is a near-hit pair and $\lambda \le 25$ against $\sqrt{P_1} = 101$: $\Delta \approx (101/25)^2 \approx 16$. This* *configuration is the unique outlier (*ratio $= 3.19$ *vs. generic* ≤ 0.08*) in the 38-configuration numerical sweep that motivated the theorem — the defect factor identifies it in advance.*

Remark 5 (honesty register)**.** *Theorem* **??** *is proved in full. In Theorem* **??** *two steps are stated at sketch level and flagged for the verification pass: the one-dimensionality/difference argument generalizing to rank* $K-1$ *(standard, but the constant's* K*-dependence must be tracked), and the successive-minima box count (Minkowski's second theorem, classical). The lemma's factorization is standard. All constants throughout are effective and integer-Diophantine; no transcendence input occurs anywhere.*